

Targets to Monitor (1 of 2) ●

- Many cloud resources generate API messages that report metrics:
 - These APIs provide specific metrics, also called **KPIs (key performance indicators)**
- Common metrics to track include the following:
 - **Overall utilization**
 - **Elasticity usage**
 - **Performance**
 - **System or service availability**
 - **Cost**

Monitoring Tools (1 of 3) ●

- Some monitoring tools support **ConMon (continuous monitoring)**
This means the tool provides near-immediate feedback on performance or security metrics in a continuous way.
- **Identifying Cloud Resources**
 - **Tags** are key-value pairs used to track cloud resources within an organization
 - To use tags effectively, a tagging policy should be created early in the cloud deployment
 - Two layers of tag design include the following:
 - **Explicit tagging**
 - **Implicit tagging**

Monitoring Tools (2 of 3) ●

- **Capturing Network Traffic**
 - A network **TAP (test access point)** captures traffic flowing between devices
 - A **packet broker** near the TAP can sort traffic, decrypt TLS traffic, mask sensitive data, and send traffic to monitoring tools
 - Common targets include a **WAF**, a **SIEM**, or an **IDS/IPS**
 - Some vendors capture cloud traffic and send it to on-prem or cloud-based packet brokers for monitoring

Monitoring Tools (3 of 3) ●

- **Monitoring User Experience**
 - **ITSM (IT service management)** tools help IT teams monitor customer-facing processes to meet user needs and expectations
 - Some core ITSM practices include the following:
 - **Service request management**
 - **Knowledge management**
 - **IT asset management**
 - **Incident management**
 - **Problem management**
 - **Change management**

Dashboards

- A well-structured dashboard

Gives a quick overview of critical services and highlights issues that may show problems.

- Tips to design an effective dashboard include the following:

- **Five-second rule**

You should identify the needed information within five seconds.

- **First things first**

Show the most critical indicators first, then trends.

- **Birds of a feather**

Group related information close to each other.

- **Less is more**

Use fewer than 10 visualizations or widgets on a dashboard.

- **A picture is worth a thousand words**

Choose visualizations carefully.

- **It's what you don't see**

Space between visualizations makes the dashboard easier to read.

Section 9-2: Events and Logs

- Two common data sources for monitoring cloud services are **events** and **logs**

Events

- An **event**

Is a detectable change in a cloud service or resource that is being monitored.

- Event collection

Happens automatically in the background.

- **Event correlation**

Finds relationships between events so tools can detect patterns or anomalies related to performance or security issues.

- Collecting too many events

Can slow down scanning processes and increase false positives.

- Event collection policies

Must define which events are tracked and how the system responds to them

Logs

- **Logs**

Collect data over time to show history, usage patterns, and anomalies.

- Three common types of logs include the following:

- **Access/authentication**

- **System**

- **Application**

- Logs must be analyzed by a service:

– Examples include AWS CloudWatch Logs Insights, Azure Monitor Logs, and GCP Log Analytics

Syslog (1 of 4) ●

- A **SIEM (Security Information and Event Management)** solution Provides log analysis to detect security and performance issues.
- Logs can be centrally collected using the syslog utility:
 - Syslog is a standard used to generate, store, and process event messages on networked systems.
- The syslog standard includes three main components:
 - **Event message format**
 - **Event message transmission**
 - **Event message handling**

Syslog (2 of 4) ●

- The syslog standard defines two roles for devices in logging events:
 - **Generator**
A device that creates and sends event information to a syslog system.
 - **Collector**
A server that receives and gathers event messages from generators.
- Syslog assigns a **severity level**
Each event is given a logging level or priority value.
- The syslog utility
Does not send alerts but keeps a history of system messages.

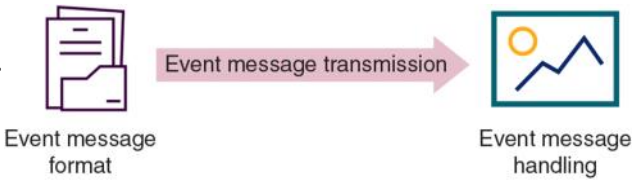


Figure 9-5 Three components of the syslog standard

Level	State indicated
0	Emergency! System unusable
1	Alert—Immediate action needed
2	Critical—Critical condition
3	Error—Error condition
4	Warning—Warning condition
5	Notification—Normal but significant condition
6	Informational—Informational message only
7	Debugging—Helpful for debugging

Figure 9-6 Syslog security levels

SNMP Communications (1 of 3) ●

- Organizations use enterprise-wide network management systems To monitor network activities in real time across the entire network.
- The following are components of a network management system:
 - **NMS (network management system) server**
Collects data from multiple managed devices.
 - **Managed device**
Any network device monitored by the NMS server.
 - **Network management agent**
Software that collects information about a device and sends it to the NMS.
 - **MIB (Management Information Base)**
A list of objects that are managed by the NMS.
- Most agents use **SNMP (Simple Network Management Protocol)**
SNMP is part of the TCP/IP protocol suite.

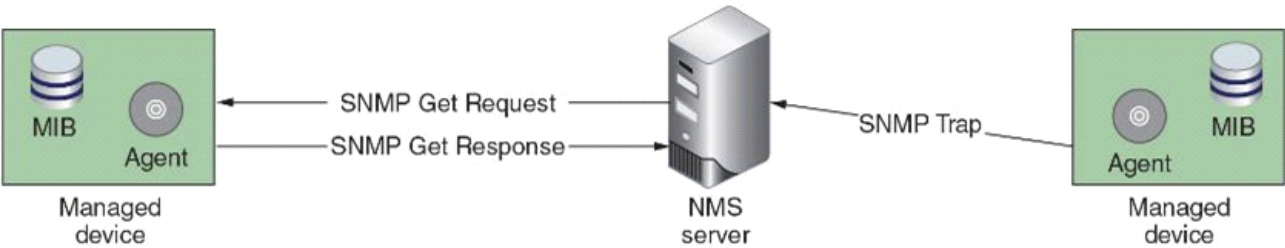


Figure 9-8 SNMP message types

SNMP Communications (3 of 3) ●

- Security measures for older SNMP versions include the following:
 - Disable SNMP on devices where it is not needed
 - Limit approved sources of SNMP messages
 - Use **read-only mode** so devices cannot be reconfigured
 - **Configure strong passwords**, called **community strings**, on SNMP devices
 - Use **different** community strings for different device types
- Cloud-native monitoring solutions
 - Are more efficient, effective, and easier to manage than older SNMP-based solutions.

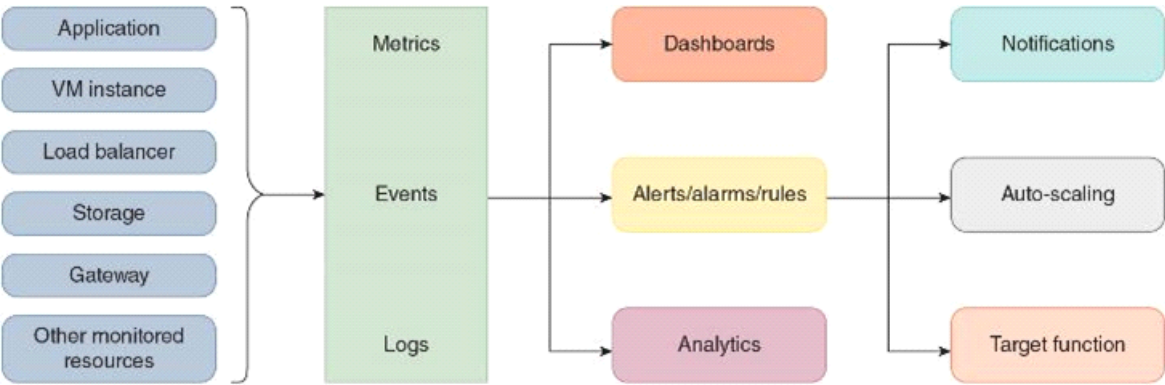


Figure 9-10 Cloud monitoring data collection and response

SNMP is old , Cloud-native monitoring is new.